

Introduction to Algorithm Engineering

Quiz Examination

March 31, 2026

Instructions:

- The quiz is for 45 minutes and 20 points.
- There are four questions in Section I and two questions in Section II. Make sure your question paper is printed correctly.
- No clarifications shall be provided during the exam. If you need to make any assumptions, state those explicitly.

Section I: Answer the following questions in brief.

1. Define the terms biconnected components and strongly connected components of an undirected graph and a directed graph, respectively. (Points:1+1=2)
2. What is meant by the inclusive property of a cache? Explain with an example. (Points:1+1=2)
3. Find the articulation points of the graph in Figure 1 using any of the algorithms discussed in class.

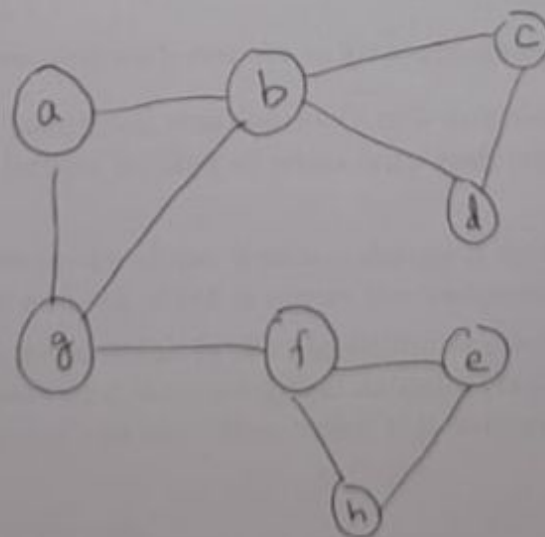


Figure 1: Graph for Question 3, Section A

(Points:2)

4. Consider the mesh graph of $n \times n$ vertices as shown in Figure 2. If we use the algorithm of Crescenzi to find the diameter of this graph, how many BFS traversals are needed? Justify your answer. (Points:2)



Figure 2: A mesh graph of size $n \times n$.

Section II: Answer each question with enough details.

1. Recall the problem of transposing a square matrix of dimensions $n \times n$. Answer the following questions.
 - (a) Write a simple algorithm for the transpose of a matrix and estimate the number of cache misses occurred.
 - (b) Write a cache-aware algorithm to transpose a square matrix.
 - (c) Estimate the number of cache misses incurred.(Points:1+2+3=6)
2. Answer the following questions with respect to strongly connected components.
 - (a) Suppose a strongly connected component C of a directed graph G has k vertices. What is the smallest and largest number of edges with each edge having both its endpoints in the component C ?
 - (b) How does the answer to the above question change if we require that only one end point is in the component and the other is not in the component C ?
 - (c) Suppose that each recursive call in the algorithm we studied in class to find the strongly connected components of a directed graph actually results in equal sized subproblems. What is the run time of the algorithm under this assumption. Justify your answer.(Points:2+2+2=6)